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Section: Original Research

Article Title: Basal Serum Cortisol and Testosterone/Cortisol Ratio are Related to Rate of Na^+ Lost During Exercise in Elite Soccer Players

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Running Head: T/C ratio associates to Na^+ lost

Journal: *International Journal of Sport Nutrition and Exercise Metabolism*

Acceptance Date: June 5, 2019

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DOI: https://doi.org/10.1123/ijsnem.2019-0129

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Rapid Communication

Basal serum cortisol and testosterone/cortisol ratio are related to rate of Na^+ lost during exercise in elite soccer players

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Abstract

During exercise, human body maintains optimal body temperature through thermoregulatory sweating, which implies the loss of water, sodium (Na^+) and other electrolytes. Sweat rate and sweat Na+ concentration show high interindividual variability, even in individuals exercising under similar conditions. Testosterone and cortisol may regulate sweat Na^+ loss by modifying the expression/activity of the cystic fibrosis transmembrane conductance regulator (CFTR). This has not been tested. As a first approximation, we aimed to determine whether basal serum concentrations of testosterone, cortisol or the ratio testosterone/cortisol relate to sweat Na^+ loss during exercise. Twenty-two male elite soccer players participated in the study. Testosterone and cortisol were measured in blood samples before exercise (basal). Sweat samples were collected during a training session, and sweat Na^+ was determined. Basal serum concentrations of testosterone and cortisol and their ratio were (mean [SD]) 13.6 [3.3] pg/ml, 228.9 [41.4] ng/ml, and 0.06 [0.02], respectively. During exercise, the rate of Na^+ loss was related to cortisol ($r=0.43$; $p<0.05$) and to the ratio testosterone/cortisol ($r=-0.46$; $p<0.01$), independently of the sweating rate. Our results suggest cortisol and the ratio testosterone/cortisol may influence Na^+ loss during exercise. Whether this regulation depends on the CFTR is unknown.

Keywords: Salty sweaters; Testosterone; Cortisol; sweat [Na^+]; Soccer; Dehydration.

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Introduction

During exercise, eccrine sweat glands secrete a hypotonic fluid (i.e., sweat) to increase evaporative cooling. The transference of heat from the body to sweat eliminates metabolic heat and thus contributes to the maintenance of core temperature (Gleeson, 1998). Sweat contains small amounts of electrolytes, in lower concentration than plasma, because eccrine sweat glands reabsorb sodium (Na^+) and chloride (Cl^-) from sweat. In healthy individuals, sweat Na^+ concentration ([Na^+]) during exercise typically ranges between 20-70 mmol/L (Baker, 2017), leading to approximately 20-90 mmol of sodium lost in one exercise session (Turner & Avolio, 2016). The mechanisms behind the interindividual variability of sweat electrolyte loss and concentration are not completely understood. However, sweat rate (Buono et al., 2007), concentration of aldosterone (Yoshida et al., 2006) and vasopressin (Hew-Butler et al., 2014), and Cystic Fibrosis Transmembrane Regulator (CFTR) might be partly responsible of this variation (Del Coso et al., 2016).

Sweat glands express CFTR, which helps to reabsorb Na^+ and Cl^- in the duct (Holguin, 2018). Upon cholinergic stimulation, the precursor fluid containing [Na^+] and [Cl^-] similar to plasma, passes through the sweat duct while electrolytes are partially reabsorbed. The efflux of Cl^- ions through CFTR causes the lumen of the duct to become electronegative with respect to the tubule cells. This situation generates an electrochemical gradient that reabsorbs Na^+ from the duct to the lumen. As a result, both [Na^+] and [Cl^-] of the precursor fluid are progressively reduced until sweat reaches the skin. A malfunction or decreased expression of CFTR (Brown et al., 2011) in sweat glands decreases sodium reabsorption, leading to abnormally high [Na^+] in sweat (i.e., > 70 mmol/L); (Del Coso et al., 2016).

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Notably, in *in vitro* and in animal models, cortisol (C) and testosterone (T) have been reported to regulate the expression of the CFTR. Cortisol downregulates CFTR expression (Laube, Bossmann, & Thome, 2015), whereas testosterone upregulates CFTR expression (Khadijah Ramli, Giribabu, & Salleh, 2018). Whether the serum concentrations of cortisol and testosterone associate with sweat Na^+ loss and concentration during exercise in humans is unknown.

In soccer, likewise in other individual and team sports, players lose considerable quantities of Na^+ through sweat during training sessions and competitions (Shirreffs, Sawka, & Stone, 2006). Previous evidence suggests sweat Na^+ losses are not replenished in soccer, due to a mismatch between sweat and rehydration rates, along with consumption of drinks low in Na^+ (Shirreffs et al., 2005). However, the identification of individual sweat Na^+ losses during soccer practices and matches might aid for a correct replenishment of Na^+ that ultimately would help in the maintaining blood volume, thermoregulation and performance (Del Coso et al., 2008; Strazzullo & Leclercq, 2014). The replenishment of the sweat Na^+ losses might be specially relevant in soccer practices performed in the heat or in those competitions where several games are played in the same day. With the aim of increasing the knowledge about the factors that influence sweat [Na^+], we used a soccer practice to determine whether basal serum concentrations of testosterone, cortisol or the ratio testosterone/cortisol were related to Na^+ and fluid losses in sweat.

Methods

Twenty-two male elite soccer players from the Chilean professional league participated in the study. Age, body characteristics, soccer experience, maximal oxygen uptake (VO_{2max}) and anthropometry are in table 1. Prior to the measurements, players underwent a pre-participation screening that included a medical and training history. Players were encouraged to avoid medications or nutritional supplements for the duration of the study. One week before the start of

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the study, players were fully informed of the experimental procedures and the risks and discomforts associated, and gave their informed written consent.

On the day of the training session, players arrived at the laboratory at 07:30-08:30 h after an overnight fast. Ten ml of blood were collected via venipuncture of an antecubital arm vein. Players then consumed a breakfast buffet. Afterwards, players went to the training facilities, emptied their bladders, and a urine sample was obtained. Immediately after, the skin was carefully cleansed with deionized water, and sweat patches were then applied in the upper back. The training session (120-145 min) was performed outdoors between 10:00 and 12:10 h. Mean ambient temperature, relative humidity, and wind speed were 32.6ºC, 25.9% and 0.7 m/s, respectively. The training session consisted of warm-up, conditioning drills (e.g., plyometrics, sprints), 45 min of a simulated soccer match (not all players completed this 45 min), and cool-down. Before and after training, all players were weighed (TanitaWB3000, USA) in dry underwear. Each player was provided with two individually numbered bottles (one of them with sports drink [Gatorade®, USA] and the other with plain water [Cachantun®, Chile]). Bottles were weighed (Camry, QE-KE-4, Korea) before and after the training session to establish the volume of fluids consumed by each player and to estimate carbohydrate and sodium intakes. Players were instructed to drink *ad libitum* only from their own numbered bottles and to not to spit out any drink. Sweat rate was determined considering the difference between post- and pre-body weight, along with the amount of liquid consumed.

Maximal Oxygen Uptake (VO_{2max})

VO_{2max} was determined with a breath-by-breath pulmonary gas exchange system (Metalyzer 3B, Cortex, Germany) during an incremental treadmill test. The starting speed was 8 km/h, with speed increments of 1 km/h every 60 s, with a fixed slope of 1.5%. VO_{2max} criteria

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were: 1) Plateau in VO_2 despite increase in workload; 2) Respiratory exchange ratio (RER) ≥ 1.10; and 3) Maximal heart rate expected for age (220 bpm - age). Tests were ended when two of those three criteria were met.

Urine Specific Gravity

Before the training session, hydration status was estimated using the urine specific gravity (USG) with a digital refractometer (ATAGO Inc, Bellevue, USA), as suggested (Castro et al. 2015).

Hormonal measurements

A teflon catheter was placed into an antecubital vein and blood samples (4 mL) were taken while sitting. Blood samples were placed in lithium-heparin collection tubes and maintained at ambient temperature for 45 min, and then centrifuged (10 min, 2000 rpm). Serum was removed and frozen at -20ºC for later analyses. Basal serum concentrations of free testosterone and cortisol were determined, in duplicate, using radioimmunoassay kits (Immunotech Beckman, Coulter Laboratories, France).

Sweat Na^+

Sweat samples were collected during the training session from the upper back using an absorbent patch applied to the skin (Tegaderm pad, 3M health care, USA). The patch was placed on the right side of the body. After the end of the training session, the patches were removed with sterile tweezers, and were immediately introduced into a 5-ml syringe. Syringe plunger was pressed to compress the patch, and the sweat was collected into a cryovial. The sweat collected was analyzed for sodium using ion electrode method (Compact [Na^+] meter B-722, Horiba, Japan)

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in accordance with the syringe protocol (Baker et al., 2014). Whole-body [Na^+] was estimated with regression equations (Baker, Stofan, Hamilton, & Horswill, 2009).

Statistics

Data were obtained as previously indicated while the testosterone/cortisol ratio was calculated by dividing the serum concentration of testosterone by the serum concentration of cortisol in each individual. Data are presented as mean [standard deviations]. Relationships were analyzed with Spearman correlation test. Linear regression models were used to compute the residuals (measured values – expected values) of relationships. Mann-Whitney U test was used for comparisons between groups. IBM® SPSS® Statistics version 25 was used for analyses. Statistical significance was set at $p \leq 0.05$.

Results

Basal (pre-training) concentrations of testosterone and cortisol and ratio testosterone/cortisol were 13.6 [3.3] pg/ml, 228.9 [41.4] ng/ml and 0.06 [0.02], respectively. Pre-training session USG was 1.021 [0.007]. Sweat loss and sweat rate were 1975 [551] ml and 0.87 [0.24] l/h, respectively. Sweat [Na^+] during training was 35 [13] mmol/L (minimum 11 mmol/L; maximum 59 mmol/L) and the rate of Na^+ loss was 0.70 [0.33] g/h (minimum 0.28 g/h; maximum 1.29 g/h).

The rate of Na^+ loss correlated directly with basal cortisol, but inversely with the ratio testosterone/cortisol (Table 2). The rate of Na^+ loss also trended ($p = 0.093$) to correlate inversely with testosterone (Table 2). Of note, sweating rate correlated inversely with testosterone and with the ratio testosterone/cortisol, and trended ($p = 0.098$) to correlate directly with cortisol (Table 2).

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We calculated the residuals of the relationship rate of Na^+ loss vs. sweating rate. Positive values indicate that the measured rate of Na^+ loss was higher than the predicted rate of Na^+ loss (based on the sweating rate); whereas negative values indicate that the measured rate of Na^+ loss was lower than the predicted rate of Na^+ loss. These residuals thus represent the rate of Na^+ loss adjusted for sweating rate. Notably, the residuals correlated with both cortisol ($r = 0.43$, $p < 0.05$) and with the ratio testosterone/cortisol (Figure 1A).

We then divided players in groups of low or high rate of Na^+ loss, independently of sweating rate and the ratio testosterone/cortisol. To that end, players with negative residuals in the relationship of Figure 1A were classified as "Low Na^+ loss"; players with positive residuals were classified as "High Na^+ loss" (Figure 1B). The rate of Na^+ loss and the sweat [Na^+] were higher in the "High Na^+ loss" than the "Low Na^+ loss" group (Table 3). No other differences were observed (Table 3).

Discussion

We investigated whether basal concentrations of testosterone, cortisol or the ratio testosterone/cortisol related to sweat Na^+ loss during exercise in elite soccer players. Both cortisol (directly) and the ratio testosterone/cortisol (inversely) associated to the rate of Na^+ loss, independently of sweat rate. These results suggest that basal levels of cortisol and the ratio testosterone/cortisol affect the rate of Na^+ loss in sweat during exercise.

Testosterone (anabolic) and cortisol (catabolic) may influence physical performance and health status of elite athletes (Slimani, Baker, Cheour, Taylor, & Bragazzi, 2017). Also, the ratio testosterone/cortisol is considered an indicator of physiological stress (Hayes, Grace, Baker, & Sculthorpe, 2015). Our current study is the first reporting a relationship between basal cortisol and

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ratio testosterone/cortisol with the rate of Na^+ loss in sweat during exercise in elite soccer players. A previous study in healthy adults showed that "salty sweaters" had a lower expression of the CFTR protein in sweat glands compared to "normal sweaters". In that study there were no differences in the expression of the epithelial sodium channel (ENaC), the main mechanism for Na^+ absorption in sweat glands (Brown et al., 2011). Importantly, the ENaC is functionally dependent on the CFTR (Reddy & Quinton, 2003), suggesting that CFTR may regulate of Na^+ absorption during exercise.

Increases in testosterone and decreases in cortisol concentrations have been shown to modify the expression and activity of the CFTR in vas deferens of male rat and in lung cells, respectively (Khadijah Ramli, Giribabu, & Salleh, 2018); (Laube et al., 2015). These observations suggest cortisol and the ratio testosterone/cortisol could regulate CFTR function in sweat glands, modifying Na^+ reabsorption. This scenario may partly explain our findings. Further studies are warranted to evaluate effects of physiological or pharmacological modifications of cortisol and the ratio testosterone/cortisol on the rate of Na^+ loss in sweat.

Average sweat [Na^+] in our soccer players was 35 [13] mmol/L, with a large range (from 12 to 59 mmol/L). Similar results were found previously in elite soccer players during a 90-min training session (30 [18] mmol/L; from 26 to 129 mmol/L) (Shirreffs et al., 2005). According to a previous classification, subjects with sweat [Na^+] lower than 30 mmol/L are considered low-salt sweaters; subjects with sweat [Na^+] between 30 and 60 mmol/L are considered typical salt sweaters; and subjects with sweat [Na^+] higher than 60 mmol/L are considered salty sweaters (Lara et al., 2016). Therefore, the players in our study could be considered typical salt sweaters.

Our results showed that the ratio testosterone/cortisol associates inversely to sweat rate and to the rate of Na^+ loss. This suggests individuals with low ratio testosterone/cortisol (high stress)

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are at a higher risk of dehydration and excessive Na^+ loss. In fact, a previous study reported that ~23% of total Na^+ loss during exercise depended on sweat rate (Holmes, Bates, Zhao, Sherriff, & Miller, 2016). Greater Na^+ loss through sweat has been related to risk of cramps in soccer (Maughan, Merson, Broad, & Shirreffs, 2004) while American football players with a history of heat cramps possess higher sweat [Na^+] than players without a cramping history (Horswill et al., 2009). However, the association between a low ratio for testosterone/cortisol and a higher likelihood of muscle cramping during exercise merits further investigation. Relationship between testosterone and testosterone/cortisol with sweat rate could be justified by previous studies that show that testosterone (Marler et al., 2003) and cortisol (Papanek & Raff, 1994) could regulated antidiuretic hormone.

We did not measure potassium in sweat or in plasma. Thus, we could not determine whether high Na^+ loss was accompanied by high potassium loss. In addition, because serum potassium concentration in sweat is similar to plasma and stays relatively constant despite changes in sweat rate, a concentration of potassium below 10 mmol/L is used as a quality control for sweat samples (Baker, 2017). If sweat potassium concentration is above 10 mmol/L, potential issues with leaching or sample evaporation may be suspected. Thus, we cannot therefore discard the possibility that high [Na^+] concentrations resulted because sweat itself leaches skin electrolytes or due to evaporation (Weschler, 2008).

The information provided by this investigation could be useful for athletes, coaches and medical staff in the development of nutritional strategies for fluid and Na^+ intake. This might be especially relevant in situations of high physiological stress, in which the ratio testosterone/cortisol might decrease, affecting the rate of Na^+ loss through sweat. Thus, the measurement of sweat Na^+ concentration and sweat Na^+ losses during training or simulated competitions might be

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recommended during periods of the season when the testosterone/cortisol ratio may change. In conclusion, this investigation shows, for the first time, that basal cortisol concentration and the ratio testosterone/cortisol are associated to the rate of Na^+ loss in sweat during exercise in elite soccer players, even after using sweat rate as a covariable. These relationships may result from the regulation exerted by cortisol and the ratio testosterone/cortisol on the CFTR protein in sweat glands. Further studies are required to test such hypothesis.

The authors declare that there is no duality of interest associated with this manuscript.

Acknowledgments:

We are grateful with all the players participating in this study.

The study was designed by JC, CP and MC; data were collected and analyzed by JC, CP, JDC, RFV and MC; data interpretation and manuscript preparation were undertaken by JC, CP, SJ, RR, HZ, JDC, RFV and MC. All authors approved the final version of the paper

The protocol was approved by the Ethics Committee of the Finis Terrae University and conforms to the policies established by the Department of Health, Education and Welfare of the United States and the American Society of Physiology. All subjects agreed to an informed consent on the effects of the study.

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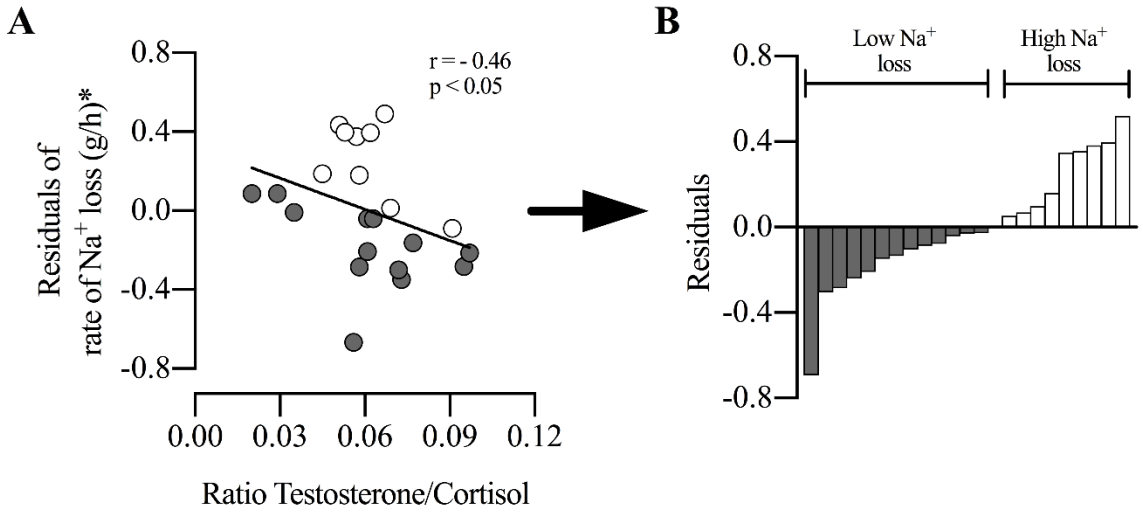

Figure 1. (**A)** Relationship between the residuals of the rate of Na^+ loss and the ratio testosterone/cortisol. *Computed from the relationship rate of Na^+ loss vs. sweating rate. (**B)** Residuals (measured – expected) of the relationship presented in A. Players with negative residuals were classified as those with low Na^+ loss. Players with positive residuals were classified as those with high Na^+ loss.

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Table 1: Characteristics of the subjects.

	Mean [standard deviation]
Age (years)	25.6 [5.8]
Height (cm)	177 [6]
Body mass (kg)	78.2 [7.5]
Body mass index (kg/m^2)	24.8 [1.8]
VO_{2max} (ml/kg × min)	55.7 [3.8]
Experience in elite level (years)	8.0 [5.2]

n = 22

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Table 2: Spearman correlations between hormone concentrations and sweat characteristics.

	Na^+ concentration (mmol/L)	Rate of Na^+ loss (g/h)	Sweating rate (L/h)
Testosterone (pg/ml)	-0.11	-0.36^A	-0.53^B
Cortisol (ng/ml)	0.38^A	0.44^B	0.36^A
Ratio Testosterone/Cortisol	-0.39^A	-0.59^C	-0.65^C

Values represent r values. $^A P < 0.10$, $^B P < 0.05$, $^C P < 0.01$, n = 22.

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Table 3: Comparison between players with low and high Na+ loss.

	Low Na^+ loss	High Na^+ loss	p value
Age (years)	24.9 [5.9]	26.6 [5.7]	0.52
Height (cm)	176 [6.5]	179 [5.1]	0.31
Body mass (kg)	76.8 [8.1]	80.3 [6.5]	0.28
Body mass index (kg/m)	24.6 [1.5]	25.1 [2.2]	0.59
VO_{2max} (ml/kg × min)	56.5 [4.4]	54.6 [2.8]	0.28
Soccer experience (years)	7.4 [5.4]	8.8 [5.0]	0.54
During exercise			
Pre USG (g/ml)	1.019 [0.008]	1.024 [0.005]	0.11
Duration of training (min)	135 [9]	137 [9]	0.60
Weight loss (kg)	-0.7 [0.7]	-0.6 [0.7]	0.74
Weight loss (%)	-0.9 [0.9]	-0.8 [0.8]	0.64
Sweat loss (mL)	1969 [616]	1982 [479]	0.95
Sweating rate (L/h)	0.9 [0.3]	0.9 [0.2]	0.97
Sweat [Na^+] (mmol/L)	26 [8]	47 [7]	<0.0001
Rate of Na^+ loss (g/h)	0.5 [0.2]	0.9 [0.3]	0.002
Water intake (mL)	645 [598]	565 [320]	0.69
Isotonic intake (mL)	600 [598]	794 [305]	0.33
Total drink intake (mL)	1246 [555]	1360 [276]	0.53
Carbohydrate intake in drink (g)	36 [36]	48 [18]	0.33
Sodium intake in drink (g)	0.27 [0.27]	0.36 [0.14]	0.33
Testosterone (pg/ml)	12.9 [3.8]	14.6 [2.3]	0.20
Cortisol (ng/ml)	219 [40]	243 [42]	0.19
Ratio Testosterone/Cortisol	0.06 [0.02]	0.06 [0.01]	0.99

Values are mean [standard deviation]. n = 13 for Low Na^+ lost; n = 9 for High Na^+ loss.

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